

Composer Book: Architecture, Interaction and Risk Controls

Four systematic engines, how each decides, how they combine, and every rule that can move capital or force cash. Prepared as a reference for correlation analysis against an external strategy.

1. The book at a glance

The account runs four independent Composer symphonies at fixed target weights. Each is a decision tree evaluated on the prior close; the winning branch determines what is held the next session. No engine reads any macro, rates, news or economic input — every condition is a price, moving-average, RSI, return or volatility statistic computed from tradeable instruments. This has been verified both by tree audit and by measurement.

Engine	Target	Role	Decision variable	Rebalance
Holy Grail (HG)	29%	Leveraged trend + dip capture	TQQQ 10/20d MA cross, TQQQ & SOXL 10d RSI	Daily
KMLM Switcher	29%	Regime switcher (long/short/duration)	10d RSI on 17 equity/vol tickers; XLK vs KMLM	Threshold
Crash Convexity Sleeve	27%	Crisis hedge / inverse + bond frontrun	Cumulative return, stdev and RSI on credit & duration ETFs	Daily
VIX Harvester	15%	Short-vol premium with credit guard	SPY 2–10d RSI; UVXY 14d RSI guard	Daily

Weights are targets, not hard bands. Drift is allowed and is only corrected by the concentration cap (Section 5), by owner instruction, or when new capital is deployed.

2. How each engine decides

2.1 Holy Grail — leveraged trend with a dip-buy branch and a cash default

A four-level tree, all conditions keyed to TQQQ (Nasdaq 3x) with one SOXL check. Read top to bottom; the first matching branch wins.

State	Condition	Holds
Uptrend, overbought	TQQQ 10d MA > 20d MA and TQQQ RSI(10) > 80	Lowest-RSI of UVXY / VIXY / VIXM (long volatility)
Uptrend, normal	TQQQ 10d MA > 20d MA	Highest-RSI of TQQQ / UPRO / UDOW / SSO / TNA
Downtrend, capitulation	10d MA < 20d MA and TQQQ RSI(10) < 30	Lowest-RSI of TECL / QLD / LABU / USD / SMH (buy the dip)
Downtrend, semis oversold	SOXL RSI(10) < 30	BIL (cash)
Downtrend, below trend	TQQQ price < 20d MA	BIL (cash)
Otherwise	—	Highest-RSI of the five leveraged trend names

Why it matters for correlation: HG is the book's highest-beta component and its only systematic equity-rebound harvester. It sits in cash roughly 23% of all days — historically these cash phases cluster in chop and have resolved upward (six prior cash-heavy stretches, forward 3-month returns positive six times out of six). Its beta to the market is therefore strongly state-dependent, not

constant.

2.2 KMLM Switcher — a three-layer regime machine

Structurally the most complex engine, and the largest single source of book return and book variance. It resolves in three stages:

Layer	Test	Result
1. Overbought scan	10d RSI > ~79 on any of QQQE, VTV, VOX, TECL, VOOG, VOOV, XLP(75), TQQQ, XLY, FAS, SPY	Hold UVXY (long volatility) — a melt-up hedge
2. Oversold scan	TQQQ RSI < 30 → TECL; SOXL < 30 → SOXL; SPXL < 30 → SPXL; LABU < 25 → LABU	Buy the dislocated leveraged sector
3. Regime default	RSI(XLK) > RSI(KMLM) = risk-on	Risk-on: lowest-2-RSI of TECL / SOXL / SVIX. Risk-off: highest-RSI of SQQQ / TLT

The engine is blended 75% with a 25% VIX sleeve (VXZ/PULS). The risk-off leg is what produced the recent rotation into TLT and then SQQQ. **Correlation note:** this engine can be long equities, short equities (SQQQ), long duration (TLT), long volatility (UVXY) or in cash within the same month — so its correlation to any external strategy is highly unstable and should never be treated as a single number.

2.3 Crash Convexity Sleeve — the hedge that pays in dislocations

Two halves. An **inverse-hold** leg that goes short equity/credit when stress thresholds trip, and a **bond frontrunner** leg that positions in duration (TMF long / TMV short) on credit and rate signals. Conditions use cumulative return, standard deviation of return, and RSI across HYG, LQD, CORP, IEF, SHY, TLT, KIE, PEJ, PSQ, LABD, TMF and TMV, with fixed thresholds at -0.33%, -1.75%, -3.5%, -4%, -6% and -7.5%. Its default resting state is BOXX (box-spread cash).

This is the engine that makes the book work. It is the only component with negative correlation to equities, and it is the reason the book's down-capture is low. Its duration exposure is holdings-determined, not view-driven: it holds TMF or TMV on about 52% of days, and the sign of its rate exposure is set entirely by which of the two it is long.

2.4 VIX Harvester — short volatility, gated by credit

Nine parallel sub-strategies, one for each SPY RSI window from 2 to 10 days, equally weighted. Each follows the same rule:

Condition	Action
SPY RSI(w) < 25 and UVXY RSI(14) ≤ 70	Hold ZVOL (short volatility) — sell panic
SPY RSI(w) < 25 and UVXY RSI(14) > 70	Hold PULS (cash) — the credit guard blocks the trade
SPY RSI(w) > 80	Hold VXZ (long mid-term volatility)
Otherwise	Hold PULS (cash)

The engine is mostly in cash: the short-vol leg carries an average weight of only 6.4%. The UVXY guard is the critical protection — it blocks just 35 of 812 short-vol days, but one of those was 6 February 2018, the -31% Volmageddon session. Tested alternatives (GARCH volatility-regime gates) all performed worse than this simple filter.

3. How they work together

The four engines are deliberately uncorrelated with each other. This is the whole design: each is individually volatile, and the combination is not.

3.1 Pairwise daily correlation

Common window 19 Apr 2023 – 31 Aug 2026, n = 845 trading days.

	HG	KMLM	SLEEVE	HARV	BOOK
HG	+1.00	+0.31	-0.11	+0.13	+0.70
KMLM	+0.31	+1.00	+0.06	+0.11	+0.84
SLEEVE	-0.11	+0.06	+1.00	+0.07	+0.24
HARV	+0.13	+0.11	+0.07	+1.00	+0.22
BOOK	+0.70	+0.84	+0.24	+0.22	+1.00

No engine pair exceeds 0.31. The sleeve is negatively correlated to Holy Grail. The book's high loadings on HG (0.70) and KMLM (0.84) simply reflect that those two carry most of the variance at 29% each.

3.2 Correlation to standard benchmarks

	SPY	QQQ	TLT	GLD	IWM
HG	+0.69	+0.67	+0.11	+0.20	+0.60
KMLM	+0.27	+0.29	-0.02	-0.03	+0.17
SLEEVE	-0.28	-0.23	-0.10	-0.06	-0.29
HARV	+0.11	+0.10	-0.02	+0.02	+0.08
BOOK	+0.47	+0.49	+0.02	+0.07	+0.35

Book correlation to SPY is 0.47 unconditionally — but that single number is misleading, and Section 3.3 is the one to use.

3.3 The number that actually matters: conditional correlation

Correlation to equities is strongly asymmetric by design. The book tracks the market on the way up and decouples on the way down.

	All days	SPY down days	SPY up days	SPY worst 5%	SPY best 5%
HG	+0.69	+0.37	+0.69	+0.19	+0.81
KMLM	+0.27	-0.01	+0.45	+0.27	+0.62
SLEEVE	-0.28	-0.32	-0.12	-0.27	-0.06
HARV	+0.11	+0.08	+0.15	+0.15	+0.02
BOOK	+0.47	+0.03	+0.65	+0.16	+0.80

Book correlation to SPY falls from 0.65 on up days to 0.03 on down days, and 0.16 in the worst 5% of sessions. That asymmetry — not the headline 0.47 — is the diversification property.

3.4 Capture ratios and behaviour by market regime

	Up-capture	Down-capture	Mean return, SPY worst 5%	Mean return, SPY down days
HG	2.15x	1.86x	-2.64%	-1.21%
KMLM	1.61x	0.13x	+0.81%	-0.09%

	Up-capture	Down-capture	Mean return, SPY worst 5%	Mean return, SPY down days
SLEEVE	-0.13x	-0.63x	+1.69%	+0.41%
HARV	0.18x	-0.10x	-0.10%	+0.07%
BOOK	1.08x	0.39x	-0.09%	-0.26%
SPY (ref)	1.00x	1.00x	-2.06%	-0.65%

The headline: on the worst 5% of market days the book averages -0.09% while SPY averages -2.06%. Up-capture is 1.08x, down-capture 0.39x. The sleeve is the engine doing that work — it earns +1.69% on exactly those days.

3.5 Correlation is not stable — read this before using any single figure

Rolling 126-day correlation between the book and SPY has ranged from **+0.09 to +0.82** over the common window. Any correlation-based allocation decision against this book should use the conditional and rolling figures rather than a point estimate. Two further warnings for an external comparison:

- **The sample is short and one-sided.** KMLM and the Harvester have only 845 trading days, all inside the 2023–2026 disinflation-and-easing regime. There is no 2022-style inflation shock in-sample. The rolling correlation between equities and rate expectations has recently turned negative (from +0.37 in January 2026 to -0.20 now), which is the configuration in which several of these relationships historically invert.
- **These are engine backtests, in-sample by construction.** The performance figures below are what the strategies did in a strong tape, not forecasts.

3.6 Standalone performance

Engine	Window	Days	CAGR	Max DD	Sharpe	Annual vol
Holy Grail	2015-06 to 2026-08	2,822	+102.8%	35.8%	1.72	47.5%
KMLM Switcher	2023-04 to 2026-08	845	+241.5%	24.5%	2.49	55.6%
Crash Sleeve	2023-01 to 2026-08	901	+39.6%	14.9%	1.51	23.8%
VIX Harvester	2023-04 to 2026-08	845	+25.4%	9.5%	2.12	11.0%
BOOK 29/29/27/15	2023-04 to 2026-08	845	+98.0%	9.8%	2.92	24.4%
SPY (reference)	2023-04 to 2026-08	845	+21.7%	18.8%	1.39	15.0%

Note the combination effect: the book's max drawdown (9.8%) is lower than every single engine's, including the 9.5% Harvester on a comparable basis, while its CAGR sits between them. That is the diversification working, and it is the central claim any correlation comparison should be testing.

4. Protections inside the engines

These are conditions in the symphony trees themselves. They fire automatically at the daily rebalance, with no human involvement, and they are the only mechanisms that move a position intraday-to-next-day.

Protection	Engine	Trigger	Effect
Trend cash default	Holy Grail	TQQQ price below its 20-day MA, or SOXL RSI(10) < 30, with no capitulation signal	Routes 100% to BIL. Active ~23% of all days.
Melt-up hedge	Holy Grail	TQQQ RSI(10) > 80 while in uptrend	Switches from leveraged equity to long volatility
Overbought vol hedge	KMLM	10d RSI > ~79 on any of eleven broad/sector tickers	Entire engine to UVXY
Risk-off switch	KMLM	$RSI(XLK) \leq RSI(KMLM)$	Rotates to SQQQ (short Nasdaq) or TLT (duration)
Credit guard	VIX Harvester	UVXY RSI(14) > 70 when a short-vol entry would otherwise fire	Blocks the trade, holds PULS. Caught the 2018 Volmageddon session.
Stress thresholds	Crash Sleeve	Cumulative-return and volatility trips at -0.33% to -7.5% across credit and duration ETFs	Activates inverse (LABD/PSQ) and duration (TMF/TMV) legs
Cash base	Crash Sleeve	No stress condition met	Rests in BOXX (box-spread cash, tax-efficient)

5. Protections at the book level

These sit outside the symphonies, in a governing policy document and a monitoring script that runs after every close. Only the operations marked **pre-authorised** may execute without a human decision.

5.1 Capital operations

Operation	Status	Trigger	Action
1. Sleeve monetisation band	Pre-authorised	Sleeve outside 7–15% of crash-exposed assets (target 10%)	Rebalance sleeve to target; proceeds to the engine furthest below its weight
2. Sleeve scale-up review	Scheduled Jan 2027	≥ 120 live days, live-vs-model correlation ≥ 0.90 , annualised gap $\geq -10\%/yr$	Raise sleeve target to 15% (band 11–20%)
3. KMLM earn-back monitor	Report-only	Correlation ≥ 0.90 , paired gap $\geq -15\%/yr$, beta in [0.9,1.1], vol ratio <1.15, live max DD <39%, sustained 2 months plus one hostile-regime month	Reports that a higher KMLM weight is statistically earned. Never trades.
5. Concentration cap	Pre-authorised	Any symphony exceeds 40% of book value	Mechanical reset of all four to 29/29/27/15

The concentration cap is the book's main structural protection against unmanaged drift. In a 55-year regime bootstrap it roughly halves tail drawdown versus never rebalancing (p95 68.6% → 37.5% on the conservative lens) at no meaningful cost to return. It carries a headroom rule: the cap must stay at least 10 points above the largest engine target, because a 40% cap over a 38% target was measured firing 5.4 times a year. Expected firing rate as configured is about 1.5 times a year.

5.2 Automated alerts (no trading authority)

Alert	Threshold	Response
Engine drawdown, tier 1	15% (12% Harvester)	Automatic live-vs-model divergence diagnostic; logged, no page

Alert	Threshold	Response
Engine drawdown, anomaly tier	HG 40% / KMLM 39% / Sleeve 20% / Harvester 12%	Human alarm. These are 55-year conservative p90 levels.
Time under water	HG 420d / KMLM 165d / Sleeve 305d / Harvester 123d	Human alarm (~1.5x each engine's historical maximum)
Book drawdown	17%	Human alarm — closes the correlated-decay blind spot
Divergence failure	Live-model correlation <0.90 or vol ratio >1.30	Escalates a tier-1 breach to a human alarm
Rotation / holdings change	Any change in any engine's holdings	Logged and reported, no action
Accident gauge	External canary composite turns RED	Triggers a divergence sweep across all four engines
Concentration	Any engine >40% of book	Fires Operation 5 above

5.3 What is never automated

Changes to symphony logic; investing in new or unapproved symphonies; liquidations; going to cash at the book level; direct single-asset trades; bank transfers; and any trade outside the pre-authorised operations. During a suspected regime break the sleeve band still executes its mechanical trim, but any discretionary harvesting beyond the band is an explicit owner decision.

6. What is deliberately absent

For an external comparison it is as important to know what does not drive this book as what does. All of the following have been built, backtested and rejected on measurement:

Rejected mechanism	Result
Market-wide trend gates on the leveraged legs	Every variant cut return in every multi-year window; the gated days are the rebound days that produce the edge
Strategy-level volatility targeting	Cut Holy Grail's CAGR from 105% to 78% with Sharpe flat — its P&L rises monotonically with its own volatility
Drawdown-exit rules	No configuration improved risk-adjusted return
Variance-risk-premium position sizing	Closed negative
GARCH volatility-regime gating and sizing	92 backtested cells; the promising variant was 82% one day and sat at the 10th percentile of random gates
Rate-hike-odds conditioning	0 of 92 overlay cells improved return, Sharpe and drawdown together; 0 improved drawdown at all
Per-instrument drift-regime gating	Worse than random gates at matched participation
Valuation-based allocation (CAPE, Buffett, forward P/E)	Signal could not beat its own inverse

The common mechanism, replicated across all eight: each engine is internally convexity-aware and earns disproportionately in its own high-stress state, so any overlay de-risks precisely the states that pay. Risk management in this book lives at the **allocation** layer — weights, the concentration cap, and the detection alerts — never as a filter on engine signals.

7. Notes for correlation analysis

- Use **Section 3.3**, not the headline 0.47. Down-day correlation (0.03) and worst-5% correlation (0.16) are the figures relevant to whether this book diversifies another strategy.
- **Treat the book as four series, not one.** The engines are close to mutually orthogonal; an external strategy may correlate strongly with one and not at all with the others. The sleeve in particular is negatively correlated to equities and to Holy Grail.
- **Expect instability.** Rolling 126-day book-SPY correlation has spanned +0.09 to +0.82. KMLM alone can be long equities, short equities, long duration or long volatility within a single month.
- **Mind the sample.** 845 common days in one favourable regime, with no inflation shock and no 2008-style credit event. Where longer history was needed, it was reconstructed synthetically and carries reconstruction error.
- **Daily data is available** for all four engine curves and the blended book over the windows in Section 3.6 if a direct correlation calculation against your other strategy is wanted.

Prepared 31 August 2026 from live symphony definitions, Composer engine backtests, and the governing policy and monitoring code. All performance figures are engine backtests measured daily mark-to-market — in-sample by construction and never forecasts. Book series is a 29/29/27/15 blend with intra-month weight drift.